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## Brac University

Semester: Spring 2026

Course Code: CSE490D

Introduction to Football Data and Analytics

Faculty: TAV



Assessment: *Final*

Duration: 1 hour 30 minutes

Date: August 19, 2026

Full Marks : 43

### Instructions:

- All questions are compulsory.
- The total score will be capped at 40, including bonus marks.

## ■ Question 1 of 6 [6 marks]

- [1 mark] Define pitch control.
- [2 marks] Why is event data alone insufficient for analyzing the movement and positioning of off-ball players?
- [3 marks] Why does the pass probability model we discussed ignore the Magnus force? What are the implications of this simplification?

## ■ Question 2 of 6 [4 marks]

There are many ways to calculate the probability of the number of goals scored in a match. For simplicity, assume each minute is an independent trial in which either exactly one goal is scored or none. For a given LaLiga season, suppose the average number of goals per match was 2.5 over 90 minutes. What is the probability that a match will end 2-2 after 90 minutes? (Assume no additional time beyond 90 minutes. Also assume each goal is equally likely to be scored by either team, independently.) Use the following formula:

$$\binom{n}{k} p^k (1-p)^{n-k}$$

## ■ Question 3 of 6 [8 marks]

- [2 marks] Define sports science.
- [4 marks] A sports scientist uses a performance evaluation form completed by a coach after a training session. Classify this data collection method along two dimensions: (i) Internal vs. External load, and (ii) Subjective vs. Objective measurement. Describe the advantages and disadvantages of this framework.
- [2 marks] What can acceleration-deceleration data tell us about a player's fitness?

## ■ Question 4 of 6 [7 marks]

Study the Python code below and answer the questions that follow.

```
def arrival_time(distance, speed):
    return distance / speed

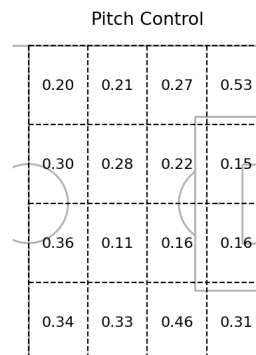
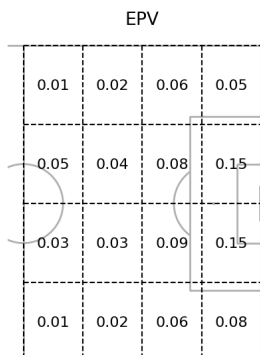
t_player1 = arrival_time(20, 8)
t_player2 = arrival_time(20, 10)

if t_player1 < t_player2:
    print("Player1_controls_ball")
else:
    print("Player2_controls_ball")
```

- [2 marks] What will be the output of this code block? Show your calculations.
- [5 marks] Why is this model an oversimplification of real-world dynamics? Propose improvements, incorporating the concepts of time-to-intercept and time-to-control in your answer.

### ■ Question 5 of 6 [10 marks]

The figures below show the Expected Possession Value (EPV) map and the Pitch Control map for the attacking team. The field is divided into  $4 \times 4$  squares. A player has passed the ball from index (2, 3) to (3, 2). [Top-left square is index (0, 0), top-right is (0, 3), bottom-left is (3, 0), bottom-right is (3, 3).]



- [2 marks] According to the expected possession value method, is this a good pass?
- [4 marks] A combined metric that incorporates both pitch control and expected possession value is sometimes used to evaluate passes. Based on this idea, rate the pass. Comment on any difference from your answer in (a).
- [4 marks] In a pitch control map, a defender is positioned far from his natural position (e.g., a left back appearing on the right side). Using the concept of “defense in attack” (defending in transition), explain how this out-of-position defender affects the team’s pitch control.

### ■ Question 6 of 6 [8 marks]

The figures below show the shot map, high expected threat (xT) passes, and cumulative expected goals (xG) timeline for a match. As a data analyst, interpret the plots to provide a comprehensive match analysis. Your answer must cover at least the Pattern of play (e.g., territorial dominance, passing direction and intent) for both teams, Match control timeline, Quality vs. volume of shots for both teams.

